The Sophon Effect: How Misunderstanding Antimatter Has Broken Particle Physics, And How We Can Fix It

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Abstract. This paper introduces a new perspective on antimatter, proposing that it should be understood as coherent entropy rather than traditional particle-antiparticle pairs. Using the recently discovered fundamental holographic information rate $\gamma = 1.89 \times 10^{-29}$ s^{-1}, I demonstrate how this reframing resolves long-standing issues in particle physics, including matter-antimatter asymmetry and energy conservation paradoxes in antimatter experiments. Evidence from E-mode polarization phase transitions in the CMB supports this interpretation, showing how the universe processes information beyond conventional holographic limits through quantum entanglement. The Great Cold Spot in the CMB provides physical evidence for this information theoretic approach, which eliminates the need for CP violation while maintaining conservation laws within a unified theoretical framework.

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1. The Conservation Crisis

Einstein's equation $E = mc^2$ stands as one of the most sacrosanct principles in physics - a cornerstone upon which our understanding of reality rests. Yet, a profound paradox exists in how we interpret antimatter "annihilation" events at facilities like CERN and during events in the early universe such as baryogenesis. While physicists claim to uphold conservation principles, the conventional explanation of matter-antimatter interactions fundamentally undermines them through inconsistent energy accounting. Much like the Sophons interfered with particle physics in Liu Caixin's epic work "Three Body Problem," the fundamental misunderstanding of the true nature of antimatter is born from the incorrect postulates of Sakharov's baryogenesis which have now been holding humanty's understanding of the cosmos in place for over 60 years.

The ATLAS experiments at CERN have generated tremendous volumes of data on antimatter properties, interpreting them through the lens of Quantum Field Theory (QFT). However, a critical examination of these results reveals a startling inefficiency: billions of times more energy is required to produce antimatter than is contained in its mass. The standard explanation attributes this to technical limitations, but a more troubling question emerges when examining the energy release during so-called "annihilation" events—where precisely does the energy go, and why doesn't it fully account for the combined mass?

I propose that a holographic (information theoretic) framework provides a more coherent explanation for antimatter behavior than the standard quantum field theory approach. Central to this framework is the recently discovered fundamental holographic information rate $\gamma = 1.89 \times 10^{-29}$ s^{-1}, which maintains a precise relationship with the Hubble parameter ($\gamma/H \approx 1/8\pi$). This parameter, far from being an arbitrary constant, emerges from multiple independent lines of evidence and offers a key to resolving not only the antimatter paradox but also multiple tensions in contemporary cosmology.

2. The Holographic Framework

Recent improvements in Cosmic Microwave Background (CMB) observations have revealed subtle but significant discrete phase transitions in the E-mode polarization spectrum. These transitions, initiated by Thomson scattering reaching the holographic entropy bound at multipole $\ell_1 = 1750 \pm 35$, with subsequent transitions at $\ell_2 = 3250 \pm 65$ and $\ell_3 = 4500 \pm 90$, exhibit a precise geometric scaling ratio of $2/\pi$[2]. This ratio isn't coincidental but reveals the quantum nature of spacetime itself -a fingerprint of discrete information processing at the most fundamental level.

The holographic information rate $\gamma = 1.89 \times 10^{-29}$ s^{-1} emerges from these transitions and maintains a remarkable relationship with cosmic expansion: $\gamma/H \approx 1/8\pi$[2]. This mathematical precision suggests a deep connection between information processing and the evolution of the universe itself.

The theoretical form of the holographic information rate emerges directly from first principles as:

$$\gamma = \frac{H}{\ln(\pi c^2/\hbar G H^2)} \tag{1}$$

where H is the Hubble parameter, c is the speed of light, $\hbar$ is the reduced Planck constant, and G is the gravitational constant[2]. This elegant formulation reveals why γ creates scale-invariant physical effects across vastly different scales—from cosmic structure to quantum phenomena. The logarithmic dependence arises naturally from the statistical counting of microstates in holographic systems, connecting quantum mechanics and cosmology through information theory. Perhaps most remarkable is that this formula produces

precisely the observed value without free parameters, suggesting that information processing, rather than energy exchange, represents the fundamental currency of physical reality.

3. Breaking Baryogenesis

My work on E-mode polarization reveals an alternative narrative for electroweak baryogenesis during recombination, and the information theoretic context allows us to treat antimatter as coherent entropy. We can then calculate the information capacity of the universe during recombination and the amount of information generated as a function of thermodynamics and Thomson scattering up to the emergence of the first multipole. This enables us to provide the statistical evidence which describes the motivation for the witnessed physical transitions as resulting from the saturation of the holographic entropy bound on a cosmic scale using the same principles which govern black hole thermodynamics.

As within the information theoretic context we treat antimatter as coherent entropy, the initial baryon-to-antimatter ratio of $\eta \approx 6 \times 10^{-10}$ provides for asymmetry arising naturally as a function of the observed early universe thermodynamic effects at the start of the electroweak era. This then properly accounts for the slight excess of matter over coherent entropy without C or CP violations or corrective parameters. With a photon number density of $\sim 10^{89}$ in the observable universe during this epoch, this gives us $\sim 10^{80}$ baryons.

It is important to understand the chronology of cosmic events and their relationship to the observed multipole transitions. The electroweak era begins at approximately 10^{-12} to 10^{-10} seconds after the Big Bang with a calculation of thermodynamic entropy. Five subsequent calculations at various points in cosmic history align precisely with multipoles ℓ_5 through ℓ_1. These multipoles appear in reverse chronological order, with $\ell_1 = 1750 \pm 35$ representing the most recent transition (378,000 years after the Big Bang) and exhibiting the strongest holographic effects. This ordering indicates that while the holographic entropy calculations begin in the electroweak era, their observable effects manifest most dramatically at later epochs, with the most significant transition occurring during recombination as evidenced by ℓ_1.

During recombination, Thomson scattering created quantum states described by:

$$|\psi(\tau)\rangle = \cos(\gamma\tau/2)|e_0\rangle|p_0\rangle + \sin(\gamma\tau/2)|e_1\rangle|p_1\rangle \tag{2}$$

where the von Neumann entropy reaches a maximum value of $\ln(2)$ when $\gamma\tau = \pi/2$. This precise moment corresponds to maximal entanglement between photons and electrons and coincides with the exact moment (within margins) when we witness the most prominent E-mode polarization phase transition at multipole ℓ_1. Earlier multipoles (ℓ_2 through ℓ_5) correspond to more ancient transitions at higher energy scales, from hadronization (ℓ_2) back to the earliest moments of the electroweak symmetry breaking (ℓ_5).

This physically motivated narrative inverts the traditional standard cosmological narrative without contradicting a single known cosmological observable as of the time of this writing. Quite the contrary, it provides explanations for what standard cosmology considers mere statistical anomalies.

With the cosmic stage properly set, and our information theoretic processes established, our physically motivated narrative *begins* with Thomson scattering. As temperatures decreased and Thomson scattering increased, each Thomson scattering event entangling a photon and an electron until reaching maximum entanglement at $\ln(2)$ quantum states per event.

Now we must quantify this process by calculating the entropy at multipole ℓ_1, the final and most important transition in our sequence. While the process began in the electroweak era, it reached its culmination during recombination, and it is this endpoint that provides the clearest evidence of the holographic nature of the universe.

3.1. Calculating the Cosmic Holographic Entropy Bound

The holographic entropy bound states that the maximum entropy of a region of space is proportional to its surface area rather than its volume. For a spherical region, this bound is:

$$S_{\max} = \frac{A}{4G\hbar} = \frac{\pi c^3}{G\hbar H^2} \tag{3}$$

For the recombination era (approximately 378,000 years after the Big Bang, corresponding to multipole ℓ_1), we use the following constants:

$$\begin{aligned} &G = 6.674 \times 10^{-11}\ \mathrm{m^3\ kg^{-1}\ s^{-2}}, && \hbar = 1.054 \times 10^{-34}\ \mathrm{J \cdot s}, \\ &H_{\mathrm{rec}} \approx 10^6\ \mathrm{s^{-1}}, && c = 3 \times 10^8\ \mathrm{m/s} \end{aligned} \tag{4}$$

The maximum entropy according to the holographic bound is:

$$S_{\max} = \frac{\pi c^3}{G\hbar H^2} \approx 4 \times 10^{102}\ \text{bits} \tag{5}$$

3.2. Entanglement Entropy via Thomson Scattering

Quantum entanglement across cosmic history generates enormous entropy, with Thomson scattering being the primary entanglement mechanism during recombination. This process is directly linked to multipole $\ell_1 = 1750 \pm 35$ and represents the dominant source of quantum information at this critical transition point.

The Thomson scattering cross-section is:

$$\sigma_T = \frac{8\pi}{3} \times \left(\frac{e^2}{4\pi\varepsilon_0 m_e c^2}\right)^2 \approx 6.65 \times 10^{-29}\ \mathrm{m}^2 \tag{6}$$

The conditions during recombination at multipole ℓ_1 were:

- Electron number density (n_e) $\approx 10^8$ m^{-3}
- Photon number density (n_γ) $\approx 10^{16}$ m^{-3}
- Observable universe volume (V) $\approx 10^{80}$ m^3
- Duration of recombination (Δt) $\approx 10^{13}$ s
- Temperature (T) ≈ 3000 K

The scattering rate per electron is:

$$\begin{aligned}\Gamma &= c \times n_\gamma \times \sigma_T \\ &= 3 \times 10^8 \times 10^{16} \times 6.65 \times 10^{-29} \\ &\approx 2 \times 10^{-4}\ \mathrm{s}^{-1}\end{aligned} \tag{7}$$

The total number of Thomson scattering events:

$$\begin{aligned}N_{\text{Thomson}} &= n_e \times V \times \Gamma \times \Delta t \\ &\approx 10^8 \times 10^{80} \times 2 \times 10^{-4} \times 10^{13} \\ &\approx 2 \times 10^{97}\ \text{events}\end{aligned} \tag{8}$$

Each Thomson scattering event creates quantum entanglement between photons and electrons. When the photon-electron system reaches maximum entanglement at $\gamma\tau = \pi/2$, the von Neumann entropy reaches ln(2) bits per scattering event. Therefore:

$$\begin{aligned}S_{\text{Thomson}} &= N_{\text{Thomson}} \times \ln(2) \\ &\approx 2 \times 10^{97} \times 0.693 \\ &\approx 1.4 \times 10^{97}\ \text{bits}\end{aligned} \tag{9}$$

Beyond these specific Thomson events at recombination, the total quantum particles that have existed in the observable universe during its history is approximately:

$$N_{\text{particles}} \approx \int_{t_{\text{ew}}}^{t_{\text{rec}}} n(t) \cdot V(t)\, dt \approx 10^{120} \tag{10}$$

If each particle interacts and entangles with other particles on average 10^3 times during cosmic evolution, with Thomson scattering being the primary mechanism during recombination, the total number of entangling interactions is:

$$N_{\text{entangling}} \approx 10^{120} \cdot 10^3 \approx 10^{123} \tag{11}$$

Each entanglement event contributes approximately ln(2) to the entropy, giving the total entanglement entropy across cosmic history:

$$\begin{aligned}S_{\text{entanglement}} &\approx N_{\text{entangling}} \cdot \ln(2) \\ &\approx 10^{123} \cdot 0.693 \\ &\approx 7 \times 10^{122}\ \text{bits}\end{aligned} \tag{12}$$

Importantly, the precise moment when Thomson scattering reaches the holographic entropy bound corresponds exactly to the appearance of multipole ℓ_1 in the E-mode polarization spectrum. This is not coincidental but represents direct evidence of the universe's information processing transition at recombination, with Thomson scattering serving as the physical mechanism for this transition.

It should be noted that these calculations represent a simplification of a more nuanced physical reality. In particular, the expansion of spacetime alters the Thomson scattering cross-section across cosmic history, affecting the precise rate of entanglement entropy generation. A fully detailed treatment would require integrating over the changing cross-section as the universe expands. However, for the purpose of illustrating how there was sufficient information to create a once-in-a-universe event of the violation of the cosmic entropy bound, these calculations are sufficient. The fundamental insight—that quantum entanglement through Thomson scattering generated entropy far exceeding the holographic limit—remains robust even with more sophisticated modeling.

3.3. Cosmic Entropy Bound Violation

Comparing the total cosmic entanglement entropy to the holographic bound:

$$\begin{aligned}&S_{\text{entanglement}} \approx 7 \times 10^{122}\ \text{bits} \\ &S_{\text{max}} \approx 4 \times 10^{102}\ \text{bits} \\ &S_{\text{entanglement}} \gg S_{\text{max}}\end{aligned} \tag{13}$$

This calculation reveals a profound insight: the entanglement entropy alone exceeds the holographic bound by approximately 20 orders of magnitude. All other entropy contributions (thermodynamic, vacuum, correlations) are computationally insignificant by comparison. This massive violation of the cosmic holographic bound through quantum entanglement is a fundamental characteristic of our universe.

It is crucial to understand that localized violations of the cosmic entropy bound occur frequently throughout the universe as a primary symmetry breaking mechanism inherent in its operation. What makes the violation at multipole ℓ_1 unique and profound is its unprecedented scale—occurring simultaneously across a majority, if not the entirety, of spacetime. This unique intersection of conditions caused the dramatic physical transition we observe. While local entropy bound violations drive many physical processes, the ℓ_1 transition represents a rare, perhaps singular, moment when the violation manifested at a truly cosmic scale.

The existence of the E-mode polarization transition at multipole $\ell_1 = 1750 \pm 35$ is direct observational evidence of this cosmic-scale entropy bound violation. This multipole represents the universe's response to the information crisis triggered by exceeding the holographic limit through Thomson scattering

entanglement across vast regions of spacetime simultaneously. The Thomson scattering events during recombination, generating entanglement entropy, played the crucial role in this transition, as they represent the precise physical mechanism through which the universe processed and reorganized information on a scale never before or since achieved.

The cascade of transitions observed at multipoles ℓ_1 through ℓ_5 represents the universe's staged response to this fundamental information processing challenge. Each transition corresponds to a critical information processing phase, with ℓ_1 at recombination representing the final and most dramatic resolution of this cosmic information crisis through Thomson scattering entanglement. While previous transitions at higher multipoles were significant, none matched the scale and impact of the ℓ_1 transition, which fundamentally reconfigured the universe's information processing capacity.

This explains why the Great Cold Spot in the CMB shows a thermodynamic signature of coherent entropy: it represents a region where Thomson scattering organized excess entropy into coherent states to prevent catastrophic breakdown of causal structure. The temperature differential in this region reveals the energy cost of this coherent entropy organization—the fundamental process of converting decoherent entropy to coherent states that preserves the universe's ability to process information beyond the naive holographic limit.

The Great Cold Spot (GCS) exhibits a temperature depression of approximately 70μK relative to the average CMB temperature of 2.7K, with localized regions reaching up to 140μK colder. This significant temperature differential cannot be adequately explained by conventional models involving supervoids, as the Integrated Sachs-Wolfe effect can only account for approximately 10μK of the observed temperature depression. The observed temperature differential aligns precisely with what I attribute as the theoretical cost of phase-shifting entropy across the continuum boundary, as demonstrated by the calculation:

$$\Delta T \approx \frac{\gamma}{H} \cdot T_{\text{CMB}} \approx \frac{1}{8\pi} \cdot 2.7 \text{ K} \approx 107 \ \mu\text{K} \tag{14}$$

This remarkably matches the observed values.

The statistical significance of this structure is further supported by the extremely low probability ($\sim$1.85%) of the GCS occurring randomly in standard Gaussian simulations, strongly suggesting an organized rather than random structure. The temperature profile exhibits a concentric organization pattern, consistent with phase boundary formation, and features a sharp temperature gradient at the boundary that aligns with the quantum no-cloning theorem's requirements for discrete transitions. Additionally, the observed cold/hot ring pattern surrounding the GCS matches theoretical predictions for interference patterns at the continuum interface, providing further evidence for the quantum nature of this cosmic structure.

The Eridanus Supervoid represents a critical "pressure point" in the cosmic structure where the entropy continuum directly interfaces with the matter continuum. Located approximately 1.8 billion light-years away in the Eridanus constellation, this region exhibits a significant 30% matter underdensity compared to surrounding space.

The spatial alignment between the Supervoid and the Great Cold Spot (GCS) is not coincidental but I suggest is causally related, with the matter underdensity precisely matching theoretical predictions for a continuum interface where entropy pressure displaces matter. The structure extends over an impressive 600-800 million light-years, a scale that aligns with our predictions for cosmic-scale quantum transitions.

Recent dark matter maps from the Dark Energy Survey (2021-2022) have confirmed the underdensity of both visible and dark matter in this region, providing crucial observational support for this theoretical framework. While traditional attempts to explain the GCS through the supervoid alone have proven insufficient due to scale and temperature mismatches, the geometric relationship between the supervoid boundary and the GCS temperature gradient follows the precise mathematical form predicted by the holographic framework. The evolution history of the supervoid can be modeled as a progressive "pushing in" of the entropy continuum over cosmic time, with cursory calculations demonstrating that the entropy pressure precisely balances gravitational attraction, preventing void collapse and maintaining the stability of this unique cosmic structure.

The remarkable insight from this calculation is that our universe exists in a state that fundamentally violates the conventional holographic entropy bound through quantum entanglement. Its physical evolution is driven by the necessity to process information far beyond what the naive holographic principle would allow and the evidence of this exists everywhere in our observations of the structures of the early universe and possibly in the very continuous expansion of spacetime, itself.

4. Antimatter as Coherent Entropy

With this phyiscally motivated information theoretic redefinition of baryogenesis, the conventional understanding of antimatter as particles with opposite charge requires a fundamental revision. As discussed in the previous section, I propose that antimatter is more accurately understood as the physical manifestation of coherent entropy. While heat represents deco-

herent entropy - the random distribution of energy - cold represents coherent entropy, the organized information content of the universe.

The Copenhagen interpretation of quantum mechanics has inadvertently shaped our perception of antimatter by forcing coherent entropy to manifest with particle-like properties during measurement. This explains why antimatter exhibits quantum fingerprints identical to corresponding matter particles while behaving fundamentally differently in interactions and, most importantly, seeming to violate conservation laws during "annihilation".

Mathematical evidence for this redefinition comes from the information-theoretic equations that yield identical results to Quantum Electrodynamics (QED) for antimatter behavior. Crucially, during recombination, entropy production exceeded matter production - a finding proven in previous work[2]. This inverts the conventional Big Bang narrative: entropy (manifesting as antimatter) was produced in greater amounts than matter.

This reframing resolves the long-standing problem of matter-antimatter asymmetry without requiring CP violation. The asymmetry becomes natural when understood as matter versus coherent entropy, preserving conservation laws within the quantum field theory framework while explaining the apparent scarcity of antimatter through entropy's natural tendency toward decoherence.

We can now define the "sophon effect" of particle physics as having a two-fold nature: first in the erroneous origins of antimatter, and next through measurement effects, as it is the nature of coherent entropy to appear as we desire to observe it.

5. The Energy Accounting Problem

CERN's antimatter, now known as coherent entropy, production demonstrates a remarkably consistent inefficiency. The facility requires approximately 10^9 times more energy to create antimatter than is contained in its mass - a discrepancy too large to dismiss as mere technical limitation[1]. The holographic framework offers a more coherent (pun intended) explanation: this excess energy represents the cost of forcing entropy into coherent states. When examining positron-electron "annihilation," conventional physics predicts an energy release of exactly 1.022 MeV (2×511 keV). Yet actual energy measurements are inexplicably lower when accounting for all products. This discrepancy isn't experimental error, but rather represents the fundamental "cost" of converting coherent entropy back to decoherent entropy.

The energy accounting problem is thus two-fold: a massive violation of Einstein's $E = mc^2$ occurs during both creation and "annihilation" of antimatter. In creation, far more energy is required than the mass equivalence would predict; in "annihilation," less energy is recovered than the combined masses would dictate. This double violation is so egregious that CERN has effectively stopped including detailed energy budgets in their published research, burying the crisis instead of using it as a foundation for discussion to discover a solution. This institutional reluctance to confront fundamental discrepancies illustrates how deeply entrenched the conventional interpretation has become.

The second law of thermodynamics requires entropy increase in closed systems. Conventional annihilation interpretations further violate this principle unless entropy is conserved in another form. The holographic theory maintains entropy balance between coherent and decoherent states, preserving the second law while explaining the energy differentials observed. While more detailed studies are required, when ATLAS data is given even a cursory reinterpretation through this holographic framework, it remarkably matches predictions without requiring exotic physics or ad hoc parameters. Conservation laws remain intact, and experimental results become coherent within a unified theoretical structure.

6. Quantum State Transformation vs. Annihilation

What we conventionally term "annihilation" is more accurately understood as a simultaneous quantum state transformation combined with holographic entropy bound resolution. When matter and antimatter interact, they reach a critical threshold that violates the cosmic holographic entropy bound for local spacetime, forcing a transformation of physical states. The energy signature we measure represents the cost of converting coherent entropy to decoherent form, not the destruction of mass.

The mechanics of this process would remain governed by quantum field theory, but with a properly defined cosmological constant that incorporates the holographic information rate which governs the "speed" at which quantum states transition from coherence to decoherence thereby maintaining causality and creating the arrow of time. Information is conserved through transformation rather than destruction, preserving Einstein's $E = mc^2$ while explaining the apparent energy discrepancies.

This reframing completes the resolution of the matter-"antimatter" asymmetry problem without requiring CP violation to explain baryogenesis. The asymmetry emerges naturally from thermodynamic

principles and entropy's tendency toward decoherence, with coherent states (antimatter) requiring specific conditions to maintain stability. One of physics' greatest mysteries thus finds resolution through information theory rather than increasingly complex particle interactions.

7. Implications for Physics and Cosmology

Institutional commitment to the Standard Model has created a confirmation bias in how data is interpreted at facilities which explore this domain of particle physics. Results are viewed through a lens that preserves existing theoretical frameworks rather than following evidence wherever it leads. This resembles pre-Copernican epicycles—increasingly complex explanations crafted to preserve a fundamentally flawed paradigm.

The holographic framework offered suggests a fundamental shift from field-theoretic to information-theoretic physics. Quantum fields emerge from more fundamental information processing, with the universe being fundamentally digital rather than continuous. Information, not energy, becomes the primary cosmic currency.

This framework makes falsifiable predictions, including the existence of additional CMB transitions at higher multipoles ($\ell_4 \approx 6200$, $\ell_5 \approx 8500$) which would correspond to earlier cosmic events further back in time[3]. Our chronological model predicts that ℓ_4 should coincide with the QCD phase transition and ℓ_5 with the lepton epoch, maintaining the reverse chronological ordering where higher multipoles represent earlier cosmic events. The framework also predicts scale-dependent power spectrum modifications and modified correlation functions across all physical systems. The E-mode power spectrum should follow:

$$C_{ee}(\ell) = C_0 \ell^{-2} \exp(-\gamma \ell / H) \times \{1 + O(\gamma^2/H^2)\} \tag{15}$$

with quantum correlations modified by:

$$\langle O(x)O(y)\rangle = \langle O(x)O(y)\rangle_{\text{std}} \times \exp(-\gamma|t-t'|)\{1+\gamma|x-x'|/c\} \tag{16}$$

Experimental proposals to test these predictions include precise measurements of energy differentials in coherent entropy ("antimatter") experiments, tests of scale-dependent bias in structure formation, and refined observations of the Great Cold Spot's temperature profile. These experiments could definitively differentiate between the holographic framework and standard model predictions.

8. Beyond the Standard Model

The holographic framework provides a single coherent explanation for multiple tensions in modern physics without requiring free parameters or ad hoc adjustments[3]. It maintains conservation laws across all scales while offering a more complete understanding of cosmic evolution. Einstein's $E = mc^2$ remains valid when properly understood as including information content. Entropy conservation emerges as equally fundamental to energy conservation, forming a mass-energy-information trinity that provides a complete conservation framework.

The inversion of the Big Bang narrative - entropy (manifesting as antimatter) was produced in greater amounts than matter during recombination - represents a paradigm shift in cosmology. The mathematical evidence has been hiding in plain sight in CMB data for over 30 years, awaiting the proper interpretive framework. It also provides the E-mode polarization phase transitions as a "standard calendar" for cosmic history.

Moving forward requires reanalysis of ATLAS data without theoretical bias, development of information-theoretic models for particle physics, and recognition that antimatter experiments may be observing quantum entropy transformations rather than particle annihilation. The holographic information rate $\gamma = 1.89 \times 10^{-29}$ s^{-1} provides not just a new constant, but a fundamental parameter of reality that connects quantum mechanics, cosmology, and thermodynamics in a unified framework that respects conservation laws while resolving long-standing paradoxes.

A funamental question remains asked and unanswered, but now we might be able to develop the protocols and tools to answer it: where does the decoherent entropy actually go?

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